Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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|---|---|---|---|---|
| Title of Proposed Technology | Development of Hydrocarbon Proton Exchange Ionomers with Tunable Gas Permeability and Enhanced Processability for PEM Water Electrolyzers | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√ ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Energy & Resources | | Chemicals | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | New/renewable energy | Hydrogen-manufacturing | High polymer materials | Material technology for energy and environmental industry purpose |
| Concept of Proposed Technology | **Concept and Background**<br>Water electrolyzers, powered by renewable energy sources, provide clean hydrogen fuel without releasing greenhouse gases. When electricity is needed, fuel cells recombine the stored hydrogen and oxygen from air by reversing electrochemical reactions, converting the chemical energy to electrical energy with water and heat as the only byproducts.<br>Proton exchange membranes (PEMs) are a critical component of many electrochemical devices such as water electrolyzers and fuel cells. Although perfluorosulfonic acid polymers (PFSAs), such as Nafion™ (Figure 1a), have been the most widely used PEM due to commercial availability, good proton conductivity, and excellent chemical stability, they are not ideal materials of choice. | | | |

The drawbacks of PFSA materials include (i) high cost, (ii) difficulty of chemical modification to tune material properties, (iii) high reactant crossover, (iv) environmental concern of the use of perfluoroalkyl substances (PFAS), known as forever chemicals, as a monomer, and (v) low glass transition temperatures.[1]

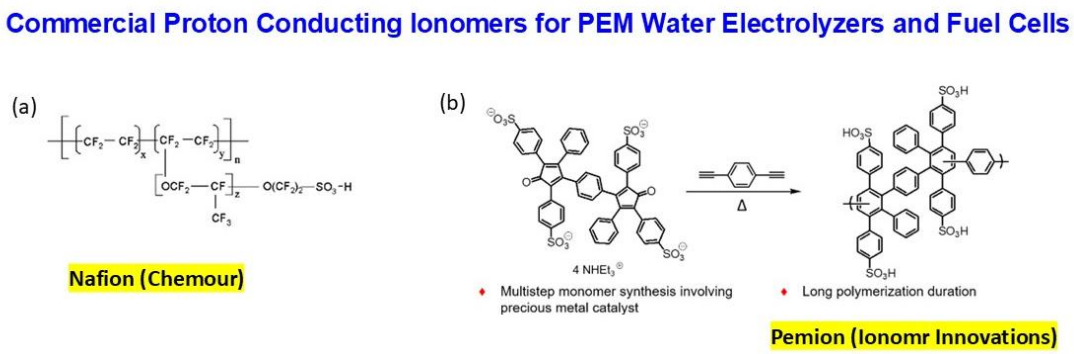


Figure 1. PEM ionomers for low temperature (<100 °C) water electrolyzer and fuel cells.

**Core technical gap (problem statement)**

To address the drawbacks of Nafion, the development of hydrocarbon PEM has been explored extensively over the past two decades.[2] Hydrocarbon PEMs based on aromatic polymers, on the other hand, can offer an alternative material solution as they are less expensive and their properties can be easily modulated, addressing the problems of current PFSA PEMs. However, hydrocarbon PEM materials generally suffer from excessive dimensional swelling in water, low proton conductivity and poor oxidative stability. Despite a sizable amount of literature existing for hydrocarbon PEMs, most of them contain aromatic polymer structures with heteroatom linkages either in backbone or sidechain.

Sulfonated aromatic polymer PEMs that are free of heteroatom linkages are supposed to afford significantly better oxidative stability in electrochemical reaction environments, but they are difficult to synthesize. For example, Holdcroft group reported improved chemical stability of sulfonated polyphenylene PEM by polymerization of sulfonated aromatics monomers as shown in Figure 1b.[3] The resulting PEM (recently commercialized as Pemion™ by Inonomr Innovations) showed excellent oxidative stability and performance; however, it has to be made by multistep synthesis of monomer preparation employing harsh conditions.

**RPI Innovation**

Recently, Bae group of RPI reported a novel, facile, and scalable one-pot synthetic method of converting an alkyl halide functionality of polymer sidechain to a sulfonate in polymer systems.[4] With this new method, sulfonated hydrocarbon PEMs can be conveniently prepared from a precursor polymer of anion exchange membranes which have recently experienced significant advances. Polyphenylene type PEMs (BPSA and *m*TPSA of Figure 2) were generated in one-pot $S_N2$ reaction of bromoalkyl sidechain of polymers followed by oxidation. These PEMs showed excellent proton conductivity with BPSA showing 250 mS/cm in water at 80 °C, nearly 1.5 times higher than that of Nafion 212. Furthermore, the separation of the sulfonic acid group from the rigid

backbone with a flexible alkyl chain mitigates excessive water uptake and in plane swelling ratio of the polymer despite of having high ion exchange capacity of 2.5 meq./g. Oxidative stability was also shown to be superior for hydrocarbon-based PEMs with negligible changes in mass, NMR, and proton conductivity.

In this TIPA collaboration project with a SME, the RPI team aims to extend our PEM work to further to lower production cost, reduce water swelling ratio, and modulate gas permeability and solvent processability. The highly tunable polymer material platform of RPI invention allows development of new proton conducting polymer materials that can be used as a membrane (i.e., PEM) and proton exchange ionomer (i.e., catalyst binder) with appropriate chemical manipulations of polymeric backbone and sidechain ionic group. The gas permeability of those novel proton exchange membrane and ionomer materials will be evaluated by a SME. Scale-up study and process optimization of down-selected polymer material will be conducted by RPI-SME collaboration.

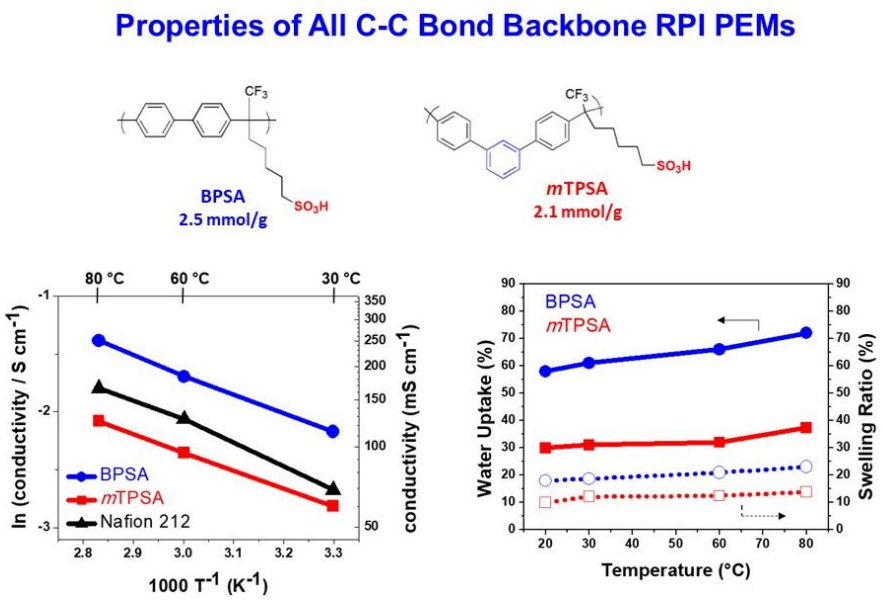


Figure 2. RPI PEMs and their membrane properties (proton conductivity, water uptake, swelling ratio in water as a function of temperature).[4]

**Rationale for Joint R&D Project**

Although PEM water electrolyzer is currently the most advanced green hydrogen production technology, only a few PFSA ionomer materials (membrane and ionomer binder) have been primarily used over the decades. Those PFSA ionomer manufactures who have dominated global market include Nafion by Chmour in the U.S., Aquivion by Solvay in the E.U., Flemion by Asahi Glass in Japan. Furthermore, as their manufacturing process is complicated (involving hazardous fluorine gas), even if demand increases, expansion of their manufacturing capability is difficult.

In this project, we propose development and scale-up study of hydrocarbon-based PEM ionomer materials which can offer alternative options to meet the growing PEM water electrolyzers and fuel cells market.

The RPI team has global reputation for high-performance PEM polymer materials development and is looking for a SME collaborator who can (a) provide complementary expertise in scale-up and

process engineering and (b) lead customer discovery and expansion in PEM water electrolyzer field. Despite the current generation RPI PEM materials have shown good performance and durability in ex situ conditions, they still suffer from several aspects for adoption in PEM industry. For example, gas permeability has been recognized as an important property in water electrolyzer, but the current RPI PEMs are not optimized to address that challenge. They still need further optimization in design and properties of PEM materials based on feedback from industry experts who can elevate TRL. Thus, multiple iterations of material development and feedback are critical for successful adoption in PEM electrolyzer system. Synergistic effect of this collaboration would result in development and commercialization of high-performance hydrocarbon PEM ionomers that satisfy various requirements for PEM water electrolyzer and fuel cell technology (Figure 3).

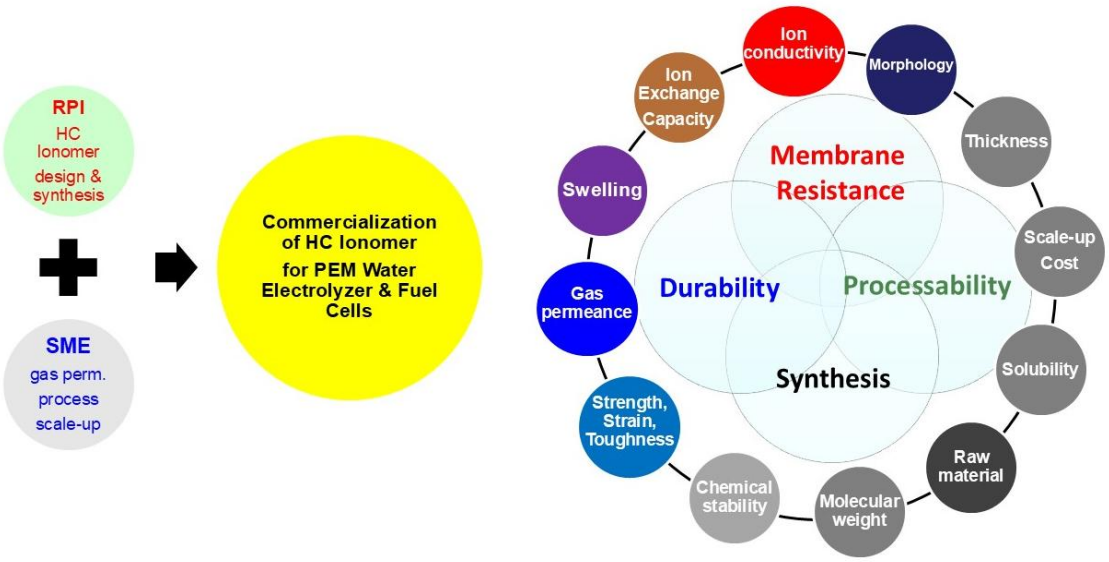


Figure 3. Integrated collaboration approach for hydrocarbon PEM ionomer development and commercialization plan that satisfy various requirements.

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|---|---|
| **Key Words** | 1. green hydrogen<br>2. water electrolyzer<br>3. fuel cell<br>4. proton exchange membrane<br>5. ionomer |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level 3 | End of the Project | Level 6 |
|---|---|---|---|---|

**R&D Objectives**

1. RPI: Develop and optimize proton exchange polymers for membrane and ionomer catalyst binder uses of PEM water electrolyzer MEAs.
2. SME: Evaluate the gas permeability and processability of RPI polymer materials for integration into MEAs and stack level. Identify shortcomings and suggest alternative materials' properties.
3. RPI and SME: Through multiple iterations of RPI materials test-and-feedback between RPI and SME, optimize materials design and synthetic process of ionomer materials for PEM water electrolyzers and establish a cost–property map and guidance on scale-up.
4. SME: Demonstrate (a) PEM ionomer production in 1 kg/batch scale and membrane casting by roll-to-roll process and (b) its commercial potential for use in PEM water electrolyzers and fuel cells.

**R&D Target Specifications**

| | Performance Indicator | Unit | Target | Global Best | Standard/ Certification Criteria |
|---|---|---|---|---|---|
| 1 | H+ conductivity (RPI) | mS/cm | 150 | 100 (Nafion) | In-plane proton conductivity in liquid water at 80 °C |
| 2 | Tensile strength (RPI) | MPa | ≥30 | 20 (Nafion) | Tensile strength measurement at 50 °C and 50% relative humidity |
| 3 | Swelling ratio (RPI) | % | ≤15% | 20% (Nafion) | Swelling ratio measurement in x- and y-direction after soaking into water for 12 h |
| 4 | $H_2$ permeance of membrane | Barrer | ≥0.7 | 0.8 (Nafion) | 80°C |
| | $O_2$ permeance of ionomer | Barrer | ≤50 | 120 (Nafion) | 80°C, 100% RH |
| 5 | Current density | A/cm$^2$ | 2.0 | 2.0 (Pemion) | 1.8V |

**R&D Plan**

| Task | Period (Month) | Key Tasks & Methodology | Main Deliverables / Milestones |
|---|---|---|---|
| Task 0 | 0–36 | Project governance (joint meeting, reporting), IP/contracting, EHS, quality management (QC) | Kick-off meting; protocol & data management plan; periodic reports; IP strategy; final technical report. |
| Task 1 SME | 0–6 | Basic property characterization of current generation RPI polymers materials (membrane and ionomer binder) for gas permeability and processability at SME for benchmarking. Optimization of processing conditions (temperature, solvent, etc). | Benchmarking report; formulation of Phase 2 plan. |
| Task 2 RPI | 3–18 | Proton exchange polymer chemistry design: select appropriate combination of backbone and sidechain ionic group for controls of molecular weight, flexibility by tuning hard/soft backbone; try different ionic groups. Characterize relationships of polymer chemical structure and mechanical property, thermal property, and membrane property (e.g., IEC, swelling). | Down-selection of 1st-gen polymer candidates (≥2); structure–property map by linking chemistry variables to membrane properties and SME feedback. |
| Task 3 | 7–24 | Benchmarking polymer synthesis and scale-up, | Interim benchmarking report, |

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| | SME | | lab scale casting, send membranes to RPI for property measurement | establishment of work plan of 4–6 |
| | Task 4 RPI | 19–30 | Optimizations of polymer synthetic process and membrane casting of down-selected 1st-gen candidates: molecular weight control, solvents selection, isolation process, membrane casting parameter, thickness optimization. | Down-select 1 polymer material candidate.<br>Delivery of thickness-optimized down-selected PEM ≥5 pieces in 10 cm x 10 cm size and ionomer binder solution ≥100 mL for lab scale demonstration. |
| | Task 5 SME | 25–36 | Scale-up of down-selected polymer (1kg/batch) and roll-to-roll process membrane casting | Final benchmarking report, Scale-up and production plan |
| | Task 6 RPI & SME | 30–36 | Scale up and commercialization planning: cost model, supply chain, production scale-up roadmap, and customer engagement for adoption. | Go-to-market plan; pilot production plan; partner/customer LOI (if available). |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | • PEM materials (membrane and ionomer) design, synthesis, and characterization: Optimize polymer architecture and synthetic chemistry to balance ion conductivity, mechanical property, and processability. |
| | • Advanced characterization of polymers by NMR, FT-IR, viscoelasticity (DMA/rheology), morphology, thermal property (DSC/SEM as needed). |
| **Expected R&R of Korean SME Partner** | • Gas permeability test of RPI PEM and ionomer.<br>• scale-up study of RPI PEM and ionomer |
| | • Identify end-use customers and expand PEM market globally.<br>• Lead commercialization: product spec, cost/production plan, and regulatory/safety documentation. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Outcomes:**<br>• Development of hydrocarbon-based PEM materials that could serve as alternative to Nafion (or other PFSAs) in global PEM water electrolyzer and fuel cell market.<br>• Optimization of multiple materials properties, manufacturing processes, and customization of PEM materials for water electrolyzer and fuel cell technologies and customers.<br>• Mapping of cost-process relationships of PEM materials and establishment of guideline of cost reduction.<br>• Establishment of PEM and ionomer production and create hydrocarbon PEM business<br>**Impact:**<br>• Establish a secure global supply chain for key materials of membrane ionomer that have potential use in global market of green hydrogen production through joint IP development.<br>• Meet the growing demand of global PEM water electrolyzer market with lower cost, more environmentally friendly material products which can accelerate green hydrogen market.<br>• Boost the global visibility and market share of Korean SMEs in green hydrogen sector by creating robust international value chains. |

## 6. References

[1] Kusoglu, A. Weber, A. Z. *Chem Rev.* **2017**, *117*, 987-1104.

[2] (a) He, G.; Li, Z.; Zhao, J.; Wang, S.; Wu, H.; Guiver, M. D.; Jiang, Z. *Adv. Mater.* **2015**, *27*, 5280-5295. (b) Adamski, M.; Peressin, N.; Holdcroft, S. *Mater. Adv.* **2021**, *2*, 4966-5005. (c) Adhikari, S.; Pagels, M. K.; Jeon, J. Y.; Bae, C. *Polymer* **2020**, *211*, 123080.

[3] Skalski, T, J. G.; Britton, B.; Peckham, T. J.; Holdcroft, S., J. *Am. Chem. Soc.* **2015**, *137*, 12223–12226.

[4] M. K. Pagels, S. Adhikari, R. C. Walgama, A. Singh, J. Han, D. Shin, C. Bae, *ACS Macro Lett.*, **2020**, *9*, 1489–1493.

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
|---|---|---|
| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |